

Nested Loops in Python

This notebook covers **only nested loops** in Python with simple explanations and examples.

1. What is a Nested Loop?

A nested loop is a loop inside another loop. The inner loop executes completely for every iteration of the outer loop.

2. Basic Syntax of Nested for Loop

In [1]:

```
for i in range(1, 4):  
    for j in range(1, 3):  
        print(i, j)
```

3. Understanding Execution Order

In [2]:

```
for i in range(1, 4):  
    print("Outer loop i =", i)  
    for j in range(1, 3):  
        print(" Inner loop j =", j)
```

4. Multiplication Table Using Nested Loops

In [3]:

```
for i in range(1, 6):  
    for j in range(1, 6):  
        print(i * j, end="\t")  
    print()
```

5. Star Pattern Using Nested Loops

In [4]:

```
for i in range(1, 6):  
    for j in range(i):  
        print("*", end="")  
    print()
```

6. Number Pattern Using Nested Loops

In [5]:

```
for i in range(1, 6):  
    for j in range(1, i + 1):  
        print(j, end=" ")
```

```
print()
```

7. Nested Loops with Lists (2D List)

In [6]:

```
matrix = [  
    [1, 2, 3],  
    [4, 5, 6],  
    [7, 8, 9]  
]  
  
for row in matrix:  
    for value in row:  
        print(value, end=" ")  
    print()
```

8. Time Complexity of Nested Loops

If both loops run n times, time complexity is $O(n^2)$.

9. Practice Tasks

1. Print a 4x4 star square
2. Print a reverse star pattern
3. Print all pairs (i, j) where i from 1–3 and j from 1–3
4. Traverse a 3x3 matrix